



Pharm*Achieve*

ACUTE AGITATION

Qualifying Exam Preparatory Course

COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ADHD Attention-Deficit/Hyperactivity Disorder

AIDS Acquired Immunodeficiency Syndrome

CNS Central Nervous System

EPS Extrapyrarnidal Symptoms

ER Extended-Release

H2 Histamine Type 2

HS Hora Somni (At Bedtime)

IM Intramuscular

IV Intravenous

ODT Orally Disintegrating Tablet

PO Per Os (Oral)

QTc Corrected QT Interval

SC Subcutaneous

SL Sublingual

TID Ter In Die (Three Times A Day)

DEFINITIONS

Acute Agitation

- A state of anxiety and restlessness that can be related to psychiatric or medical causes

Aggression

- Behaviour leading to self-assertion
- Aggression can occur with acute agitation

Delirium

- An acute, usually reversible, disturbance in mental status resulting in confusion, agitation, and fluctuating levels of consciousness
- Patients may have trouble with attention and cognition (memory deficit, disorientation, language, perception)
- The disturbance develops over a short period of time (hours to days)
- FIND ME mnemonic used to identify cause of delirium: Functional (psychiatric), Infectious, Neurologic, Drugs, Metabolic, Endocrine

CLINICAL PRESENTATION

- Angry demeanor, anxiety, confusion
- Loud/aggressive speech
- Provocative behaviour
- Tense posturing (e.g., gripping arm rails tightly, clenching fists)
- Pacing or frequently changing body position

Acute agitation can also be associated with

- Delirium stemming from any cause (e.g. electrolyte disturbances, irregular blood sugar levels, drug toxicity, infections)
- Drug withdrawal
- Neurological conditions
- Psychiatric disorders
- Dementia

Call the police right away if:

- Acts destructively (i.e. hitting things or destroying equipment)
- Is armed
- Makes verbal or physical threats that you do not have the resources (security) to contain



RISK FACTORS AND DRUG CAUSES

Risk Factors

- Pre-existing psychiatric conditions
- Personal history of violence
- Substance abuse (e.g. alcohol, stimulants)
- Substance withdrawal (e.g. CNS depressants)
- Social stressors (e.g. divorce, job loss)
- Chronic pain
- Grief
- Young age
- Male
- Poor communication among staff
- Lack of privacy

Drug Causes

- Anticholinergics
- Opioid use/toxicity/withdrawal
- Benzodiazepine withdrawal
- Corticosteroids
- Amantadine
- Dopamine or dopamine agonists
- Amphetamine abuse
- Histamine H2-receptor blockers



A diagnosis is based on clinical presentation and history

OTHER CAUSES - AGITATION

Toxicological

- Alcohol intoxication or withdrawal
- Stimulant intoxication
- Other medications (e.g. anabolic steroids, sedatives)

Neurological

- Stroke
- CNS infection
- Seizure
- Dementia
- Intracranial lesion

Medical

- Hyperthyroidism
- Hypoglycemia
- Hypoxia
- Shock
- Serotonin syndrome
- Hypothermia; hyperthermia

Psychiatric

- Psychosis
- Schizophrenia
- Mania
- Paranoid delusions
- Personality disorder

GOALS OF THERAPY

- ✓ Identify and treat (where possible) the underlying cause
- ✓ Relieve agitation
- ✓ Prevent recurrence of symptoms
- ✓ Prevent patient harm
- ✓ Prevent healthcare provider/bystander harm

NON-PHARMACOLOGICAL MEASURES

10 Key Concepts of Verbal De-Escalation

1. Respect personal space
2. Do not be provocative
3. Establish verbal contact
4. Use concise, simple language
5. Identify feelings and desires
6. Listen closely to what the patient is saying
7. Agree or agree to disagree
8. Lay down the law and set clear limits
9. Offer choices and optimism
10. Debrief the patient and staff

Clinicians must pay attention to safety when working with acutely agitated patients:

- Patients should be spoken to in a **calm environment** (quiet, low lighting)
- Patients should be **assisted in controlling their behaviour** and should be reassured that they are in a safe environment

Limit sedative use in patients with dementia

- Use non-pharmacological therapies

Adapted from: Consensus Statement of the American Association for Emergency Psychiatry Project BETA De-escalation Workgroup

S

Support

A

Acknowledge

V

Validate

E

Emotion Naming

PHARMACOLOGICAL MEASURES - AGITATION

The 3 major classes used in acute agitation are:

Benzodiazepines (most common: PO/IM lorazepam, midazolam)	First-generation antipsychotics (most common: haloperidol)	Second-generation antipsychotics (most common: ODT/IM olanzapine, risperidone, ziprasidone)
Preferred for agitation from: • CNS stimulant intoxication • Alcohol withdrawal • Undifferentiated agitation (without symptoms/history of psychosis) • In violent patients • IM options: midazolam, lorazepam • Monitor for respiratory depression • Avoid in patients with alcohol intoxication → risk of respiratory depression	Preferred for agitation with: • A known psychiatric disorder (worse side effect profile) • CNS depressant intoxication (alcohol) • In violent patients • Undifferentiated agitation (with symptoms/history of psychosis) • IM options: haloperidol, loxapine • Long history of use and experience in agitation → IM haloperidol is the standard of care. Adverse effects include EPS and QTc prolongation	Preferred for agitation with: • A known psychiatric disorder (better side effect profile) or cooperative patients or undifferentiated agitation (with symptoms/history of psychosis) • Can use in patients who develop psychosis from amphetamine use (in addition to treating with a benzodiazepine) • Could use in CNS depressant intoxication (alcohol) but less evidence • IM options: olanzapine • Newer agents with less experience outside of psychiatric settings, many industry-run trials • PO Risperidone + lorazepam appears as effective as IM haloperidol + lorazepam • Risperidone is the best studied second-generation antipsychotic in agitation • IM olanzapine has less EPS than haloperidol but more sedating and more expensive • Quetiapine causes significant orthostatic hypotension → not preferred in acute/emergency room settings • Ziprasidone causes significant QTc prolongation • Most commonly used agents are olanzapine and risperidone

PHARMACOLOGICAL MEASURES

Agitation

- **Severely violent patient:** first-generation antipsychotic + benzodiazepine
- **Violent patient:** benzodiazepine or first-generation antipsychotic
- **Intoxication with CNS Stimulant:** benzodiazepine
- **Intoxication with CNS Depressant:** first-generation antipsychotic preferred over second-generation antipsychotic
- **Patient with known psychotic/psychiatric disorder:** second-generation antipsychotic preferred over first-generation antipsychotic due to side effect profile
- **Cooperative Patient:** second-generation antipsychotic preferred

***Extreme violence:
1st generation
antipsychotic +
benzodiazepine**

Delirium

Benzodiazepines (most common: PO/IM lorazepam, midazolam)

- Preferred if cause of delirium is benzodiazepine withdrawal or alcohol withdrawal.
 - If not due to these causes, do not use benzodiazepines as they can worsen delirium
- Monitor for respiratory depression

First and second-generation antipsychotics

- Preferable to use second-generation due to more **favorable side effect profile**

PHARMACOLOGICAL MEASURES - GENERAL PRINCIPLES

- Do not administer IM olanzapine with parenteral benzodiazepines due to cardiac and respiratory complications and risk of death
- Combination therapy with oral benzodiazepines and first generations antipsychotics may be more effective → “5 and 2” means 5 mg of haloperidol and 2 mg of lorazepam (both IM or IV) in cases of severe agitation including violence

When choosing between first and second-generation antipsychotics, consider

- Availability of drugs in setting (hospital formulary)
- Dosage form: Intramuscular (IM) injection, Orally Disintegrating Tablet (ODT), Intravenous (IV)
- Concomitant medication use (if already on antipsychotics)
- Patient profile (comorbidities, cooperation level, etc.)
- Adverse effect profile (QT prolongation, cardiac risk factors)
- Onset of action (faster onset with benzodiazepines compared to IM antipsychotics)

PHARMACOLOGICAL MEASURES - GENERAL PRINCIPLES

Patients with dementia

- Lowest effective dose for shortest possible duration as these agents can worsen cognition, increase risk of cerebrovascular events and death
- Only when the benefits outweigh the risks
- **Aripiprazole, brexpiprazole and risperidone are recommended for severe agitation**
- If symptoms are refractory, quetiapine or typical (first-generation) antipsychotics may be considered
- Long-term antipsychotic use is associated with a higher risk of stroke and death
- **Cautiously titrate**
- Benzodiazepines can cause paradoxical worsening of delirium and agitation and cause prolonged sedation

Mental Health Patients (mania and psychosis)

- Treatment involves short-term control and long-term maintenance treatment

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Agitation	Improved	Minutes to hours
Vital Signs		

Safety Endpoints		
Parameter	Degree of Change	Timeframe
Side effects from medication therapy (e.g. confusion, respiratory depression, QT prolongation)	Prevent	Duration of therapy
Recurrent Symptoms		

TAKEAWAY

- **Acute Agitation:** a state of anxiety and restlessness that can be related to psychiatric or medical causes. Associated with delirium and aggression
- **Causes:** neurological, toxicological, psychiatric, medical and drug causes
- Safety of agitated patient and those around them including other patients and staff is paramount
- **Non-pharmacological therapy** such as verbal de-escalation is **first-line** where possible
- **Drug therapy** includes **benzodiazepines and antipsychotics**
- Many factors to consider when choosing between first and second-generation antipsychotics
- Long-term antipsychotic use in patients with dementia is associated with a higher risk of cerebrovascular events and death. **Use the lowest possible dose for the shortest possible duration**
- **Severe agitation:** haloperidol with lorazepam (IM injections)
- **Do not use IM olanzapine + parenteral benzodiazepines**

CASE 1

PT is a 28-year-old male who was brought in by police, who found him in a physical altercation with a bottle of vodka in his hand. Upon presentation in the ER, he was verbally aggressive with the nursing staff, unsteady on his feet, and slurred his words. After attempts to de-escalate with non-pharmacological measures, he struck the pharmacist walking in front of him. PT has a medical history of alcohol use disorder, Attention-Deficit/Hyperactivity Disorder (ADHD) and type 1 diabetes mellitus. He currently takes methylphenidate ER 27 mg PO daily, Lantus® (insulin glargine) 30 units SC HS and Novorapid® (insulin aspart) 6 units SC TID before meals. He lives in social housing and works as a dishwasher at a local restaurant. PT recently lost his close friend in an accident. He has no known drug allergies.

1. What could be contributing to PT's current agitation?
 - a) Alcohol use
 - b) Possible hypoglycemia
 - c) Stimulant use
 - d) All of the above



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2. Which medication would be the best choice for PT?
- a) Lorazepam SL
 - b) Haloperidol IM
 - c) Midazolam IM
 - d) Non-pharm measures



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3. Which of the following is NOT a side effect of haloperidol?
- a) Extrapyramidal symptoms
 - b) QTc prolongation
 - c) Sedation
 - d) Mania



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4. Which of the following agents is the LEAST appropriate for a patient with agitation due to psychosis?
- a) Risperidone
 - b) Haloperidol
 - c) Lorazepam
 - d) Ziprasidone



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CHANGE LOG

August 2025

- Formatting and grammar changes throughout
- Slide 3: Updated legend
- Slide 13: Added quetiapine recommendation for refractory symptoms
- Slides 16-19: Added route of administration to drug doses in case study
- Slide 20: Updated references

November 2025

- Slide 18: Updated question and changed option D in answer choices

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 6: Minor update to drug causes
- Slide 10: Removed ziprasidone from IM option as not available in Canada except as oral formulation
- Slide 11: Minor update to benzodiazepine use in delirium
- Slide 14 – reformatted efficacy parameters table